Crouzeix: Folie à Deux

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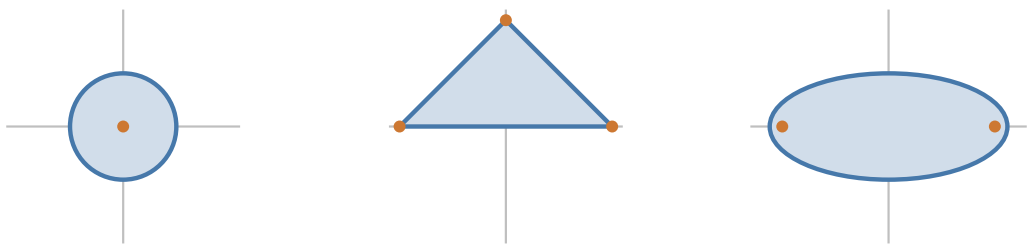
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Numerical range

Def. Let $\mathcal{H}$ be a complex Hilbert space and $T \in \mathcal{B}(\mathcal{H})$ a bounded operator. The numerical range of T is

$$W(T) = \{\langle Tx, x\rangle : \ x \in \mathcal{H}, \ \|x\| = 1\}.$$

In finite dimension closed and convex (Toeplitz–Hausdorff). In infinite dimension convex, $\overline{W(T)} \supseteq \operatorname{conv} \sigma(T)$.



$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ $\quad \operatorname{diag}(1, i, -1) \quad$ $\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$

Orange: $\sigma(T)$. Blue: $W(T)$. Normal: $\operatorname{conv} \sigma(T)$. 2×2: ellipse, foci at the eigenvalues.

Spectral sets

Def. A set $S \subseteq \mathbb{C}$ is called a K-spectral set for an operator $T \in \mathcal{B}(\mathcal{H})$ if

$$\|p(T)\| \leq K \sup_{z \in S} |p(z)| \qquad \forall\, p \in \mathbb{C}[z].$$

(S bounded, polynomially convex.)

von Neumann's inequality. The disc is a spectral set for any contraction:

$$\|p(T)\| \leq \sup_{z \in \mathbb{D}} |p(z)|, \qquad \|T\| \leq 1.$$

(Also the closed half plane if $\operatorname{Re} T = \frac{T+T^*}{2} \geq 0$.)

Two proofs of von Neumann

von Neumann	Sz.-Nagy
function theory	dilation theory
Blaschke products	unitary dilation of T
Möbius maps by hand	$p(T) = P_{\mathcal{H}}\, p(U)\big\vert_{\mathcal{H}}$

Crouzeix conjecture

Conjecture (Crouzeix 2004)

Let T be a bounded operator on a Hilbert space $\mathcal{H}$. The numerical range $\overline{W(T)}$ is a 2-spectral set for T.

Remark. This conjectured bound is sharp:

$$W\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = D(0, \tfrac{1}{2}), \qquad p(z) = z,$$

$$\left\| \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right\| = 1 = 2 \cdot \tfrac{1}{2} = 2 \sup_{D(0,1/2)} |p(z)|.$$

Antecedents

- von Neumann: constant 1, only for contractions.
- Okubo–Ando 1975, Berger–Stampfli 1967: the disc is a 2-spectral set.
- Via 2-dilations: $W(T) \subset \overline{\mathbb{D}}$ iff $T^n = 2V^*U^nV$ (Berger).
- Crouzeix 2004: true for 2×2, equality only when $W(T)$ is a disc.

Constants

Ω bounded convex, smooth boundary, $W(T) \subset \Omega$.

$$C_\Omega := \sup\left\{\|f(T)\| : \ W(T) \subset \Omega, \ \|f\|_\infty \leq 1\right\}.$$

Delyon–Delyon	1999	$C_\Omega < \infty$	depends on Ω
Crouzeix	2004	2	only for 2×2
Crouzeix	2007	11.08	uniform in Ω
Crouzeix–Palencia	2017	$1 + \sqrt{2}$	completely bounded
Jin	27 July 2026	2	function theory
Lorist–Schwenninger	Aug 2026	2	dilation theory

Technique (all of the above): the double layer potential.

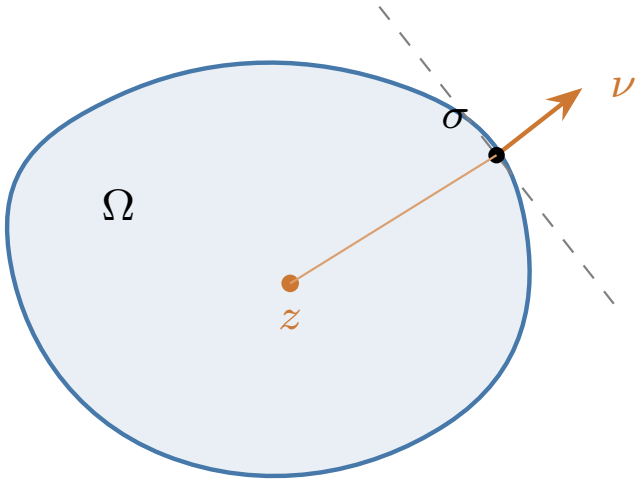
Double layer potential

$\Omega \subset \mathbb{C}$ open bounded convex, smooth boundary, $\overline{W(T)} \subset \Omega$. $A(\Omega)$: holomorphic on Ω, continuous on $\overline{\Omega}$. $C(\partial\Omega)$: continuous on $\partial\Omega$. Sup norms. For $\sigma \in \partial\Omega$ let $\nu = \nu(\sigma)$ be the outward pointing normal; for $z \in \Omega$ the double layer potential is

$$\mu(\sigma, z) = \frac{1}{2\pi}\Big(\underbrace{\frac{\nu}{\sigma - z}}_{\text{first layer}} + \underbrace{\frac{\bar{\nu}}{\bar{\sigma} - \bar{z}}}_{\text{second layer}} \Big)$$

$= \frac{1}{\pi} \operatorname{Re} \frac{\nu}{\sigma - z}$. First layer: the Cauchy kernel.

Double layer potential



- $\mu\, ds = \frac{1}{\pi}\, d\arg(\sigma - z)$: the angle at which z sees $d\sigma$.
- $\mu \geq 0$: z inside every supporting line.
- $\int_{\partial\Omega} \mu\, ds = 2$ for $z \in \Omega$; 1 for $z \in \partial\Omega$.

Why use? The Cauchy layer leads to a bound on $f(z)$. If we can control the conjugate we win. (Adjoint under the functional calculus.)

Double layer potential, cont.

Let

$$C(f,T) = \frac{1}{2\pi i}\int_{\partial\Omega} f(\sigma)(\sigma - T)^{-1}\, d\sigma \in \mathcal{B}(\mathcal{H})$$

and

$$S(f,T) = C(f,T) + C(\bar{f},T)^* \in \mathcal{B}(\mathcal{H}).$$

By definition S is the function calculus integration against the double layer potential $\mu(\sigma, z)$, i.e.

$$S(f,T) = \int_{\partial\Omega} f(\sigma)\,\mu(\sigma,T)\,ds,$$

with $\mu(\sigma,T) = \frac{1}{\pi}\operatorname{Re}[\nu(\sigma)(\sigma - T)^{-1}] \geq 0$, $\int \mu\, ds = 2I$.

The question

Lemma (Crouzeix–Palencia 2017)

For $f \in C(\partial\Omega)$,

$$\|S(f,T)\| = \|f(T) + C(\bar{f},T)^*\| \le 2\|f\|_{\partial\Omega}.$$

Proof: $\mu \ge 0$, $\int \mu\, ds = 2I$, Cauchy–Schwarz.

So the question we must answer is: how does $\|S(f,T)\|$ control $\|f(T)\|$? In particular, can we estimate $C(\bar{f},T)^*$ or make it vanish all together?

Positivity and the double layer formula

Consider

$$2\Phi(f) := f(T) + C(\bar{f}, T)^* = f(T) + \alpha(f)(T)^*,$$

where

$$\alpha(f)(z) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{\overline{f(\sigma)}}{\sigma - z}\, d\sigma.$$

Note that $\alpha : A(\Omega) \to A(\Omega)$ is antilinear and $\alpha(1) = 1$.

Positivity of Φ says $2\|\Phi(f)\| \leq 2\|f\|_{\partial\Omega}$: the result from the previous slide, found in Crouzeix–Palencia.

Positivity and the double layer formula

Scalar case: $2\Phi(f)(z) = f(z) + \overline{\alpha(f)(z)}$, holomorphic plus anti-holomorphic.

Disc: $\alpha(f) = \overline{f(0)}$, a constant.

Crouzeix–Palencia: estimate the companion, $\|\alpha(f)\| \leq \|f\|$. Gives $1 + \sqrt{2}$.

An alternate approach (not taken by either ChatGPT proof) is to "hack" the formula by studying extremal functions: f extremal, $\|f(T)\| \leq \|2\Phi(f)\|$, which proves Crouzeix. (Bickel, Gorkin, Greenbaum, Overton.)

Equivalent formulation

Theorem (Twisted real part)

Let $\mathcal{A}$ be a unital commutative operator algebra, let $\theta : \mathcal{A} \to \mathcal{B}(\mathcal{H})$ be a bounded homomorphism and let $\alpha : \mathcal{A} \to \mathcal{A}$ be a unital antilinear map such that the unital linear map

$$\theta_\alpha : \mathcal{A} \to \mathcal{B}(\mathcal{H}), \qquad a \mapsto \tfrac{1}{2}\big(\theta(a) + \theta(\alpha(a))^*\big)$$

is contractive. Then $\|\theta\| \leq 2$.

- Ransford–Schwenninger (2018): twisted real part $\Rightarrow$ Crouzeix.
- Michael Hartz (2026) proves the theorem using Jin's ChatGPT solution.

Function theory

Function theory

Theorem (Holomorphic $\to$ resolvent transfer; Jin, operator form Hartz)

Let $T \in \mathcal{B}(\mathcal{H})$ with $\sigma(T) \subset \overline{\mathbb{D}}$. Let $H : \mathbb{D} \to \mathcal{B}(\mathcal{H})$ be a holomorphic function such that $H(0) = I$ and $\operatorname{Re} H(z) \geq 0$ for all $z \in \mathbb{D}$. Assume that

$$H(z) - (I - zT)^{-1}$$

commutes with T^* for all $z \in \mathbb{D}$. Then $\|T\| \leq 2$.

$H(z) - (I - zT)^{-1} = 0$: $\operatorname{Re}(I - zT)^{-1} \geq 0$ iff $W(T) \subset \overline{\mathbb{D}}$ (Berger); conclusion is Okubo–Ando.

Transfer $\Rightarrow$ twisted real part (Hartz)

Let $a \in \mathcal{A}$, $\|a\| \leq 1$, $T = \theta(a)$. For $z \in \mathbb{D}$ the Cayley transform $c_z = (1+za)(1-za)^{-1}$ has $\operatorname{Re} c_z \geq 0$, and θ_α is positive, so

$$H(z) := \theta_\alpha(c_z)$$

is holomorphic with $H(0) = I$ and $\operatorname{Re} H(z) \geq 0$. Since θ is a homomorphism, $\theta(c_z) = 2(I - zT)^{-1} - I$, hence

$$H(z) - (I - zT)^{-1} = \tfrac{1}{2}\big(\theta(\alpha(c_z))^* - I\big),$$

the adjoint of an element of $\theta(\mathcal{A})$, which is commutative and contains T. So $H(z) - (I - zT)^{-1}$ commutes with T^*, and the transfer theorem gives $\|\theta(a)\| \leq 2$.

Proof of the transfer theorem, I

$R(z) = (I - zT)^{-1}$, $\Delta = H - R$. On $H^2 \otimes \mathcal{H}$ ($\mathcal{H}$-valued power series), $\operatorname{Re} H \geq 0$ gives $\operatorname{Re} M_H \geq 0$, and $M_H = M_R + M_\Delta$.

For W with $\sigma(W) \subset \mathbb{D}$, the functions vanishing at W:

$$N_W = \{F = \sum F_n z^n : \ \sum W^n F_n = 0\},$$

and $\Gamma : \mathcal{H} \to N_W$ its reproducing kernel at 0.

- If $\Delta(z)$ commutes with W, then $M_\Delta N_W \subset N_W$.
- Since $\Delta(0) = 0$, $\Gamma^* M_\Delta \Gamma = 0$.

Take $W = \frac{1}{2}T^*$. Then

$$\Gamma^*(M_R + M_R^*)\Gamma = \Gamma^*(M_H + M_H^*)\Gamma \geq 0.$$

Proof of the transfer theorem, II

Explicitly $\Gamma = K_0 - K_W P^{-1}$, with K_W the kernel map at W,

$$P = K_W^* K_W = \sum_{n \geq 0} 4^{-n} T^{*n} T^n,$$

$$Q = K_W^* K_{T^*} = \sum_{n \geq 0} 2^{-n} T^{*n} T^n.$$

The resolvent identity turns the last inequality into

$$4Y - YP - PY \geq 0, \qquad Y = Q - P \geq \tfrac{1}{4} T^* T.$$

If $Px = \alpha x$ with $\alpha > 2$, testing on x gives $(4 - 2\alpha)\langle Yx, x\rangle \geq 0$, so $Yx = 0$, so $Tx = 0$, so $Px = x$. Hence $P \leq 2I$. With $P = I + \frac{1}{4} T^* P T$,

$$T^* T \leq T^* P T = 4(P - I) \leq 4I.$$

Dilation theory

Dilation theory

Lemma (Lorist–Schwenninger, ChatGPT)

Let $\mathcal{H}$ be a Hilbert space and $T \in \mathcal{B}(\mathcal{H})$. Suppose there exists a Hilbert space $\mathcal{K}$, a contraction $Q \in \mathcal{B}(\mathcal{K})$ and an isometry $V \in \mathcal{B}(\mathcal{H}, \mathcal{K})$ such that the operators

$$E_n := 2V^* Q^{*n} V - T^{*n}, \qquad n \in \mathbb{N},$$

are uniformly bounded and commute with T. Then $\|T\| \leq 2$.

$E_n = 0$: 2-dilation of T^*, $\|T\| \leq 2$.

Lemma ⇒ Crouzeix

Let $\mathcal{K} = L^2(\partial\Omega, ds; \mathcal{H})$ and $Vx = \frac{1}{\sqrt{2}}\mu(\cdot, A)^{1/2}x$; V is an isometry because $\int \mu \, ds = 2I$. For $\|f\|_\infty = 1$ let Q be multiplication by f, a contraction. Then

$$V^*Q^nV = \tfrac{1}{2}\int f^n \mu \, ds = \Phi(f^n),$$

and the identity for f^n, adjointed, gives with $T = f(A)$

$$E_n = 2\Phi(f^n)^* - f(A)^{*n} = \alpha(f^n)(A).$$

E_n commutes with $f(A)$ (both come from the functional calculus) and $\|E_n\| \le \|\theta\|\|\alpha\|$. The lemma gives $\|f(A)\| \le 2$. The bound $\|\alpha\| \le 1$ is not used.

Proof of the lemma, I

Suppose $\kappa = \|T\| > 1$ and take a unit x with $T^*Tx = \kappa^2 x$ (ultrapower). Put $m_n = \mathrm{Re}\langle E_n T^n x, x\rangle$ and

$$u = Q^*VTx - \kappa V x,$$

the failure of VTx to be a κ-eigenvector of Q^*.

From the identity for $n = 1$, V isometric and Q contractive:

$$\|u\|^2 \leq 2\kappa^2 - \kappa^3 - \kappa m_1.$$

This alone gives $\kappa \leq 1 + \sqrt{1 - m_1}$; $m_1 \geq -1$ gives $1 + \sqrt{2}$, and $m_1 < 0$ for some 3×3 matrices.

Proof of the lemma, II

From the identities for all n and $E_n T = T E_n$, completing a square,

$$\kappa m_n - m_{n+1} \geq -\frac{\|u\|^2}{\kappa(\kappa-1)} \qquad (n \geq 1).$$

Multiply by κ^{-n+1} and sum; the left side telescopes:

$$\kappa m_1 \geq -\frac{\|u\|^2}{(\kappa-1)^2}.$$

With the upper bound,

$$\|u\|^2\big(1-(\kappa-1)^{-2}\big) \leq \kappa^2(2-\kappa),$$

impossible for $\kappa > 2$.

Comparison

	function theory	dilation theory
positivity	$\operatorname{Re} H \geq 0$	$\Phi = V^* \pi(\cdot) V$
all powers	$H(w) = I + 2\sum_m w^m \Phi(f^m)$	E_n, all n
companion	$\sum_m w^m \alpha(f^m)(T)^*$	$\alpha(f^n)(T)$
used	commutes with T^*	commutes with T
removal	$\Gamma^* M_\Delta \Gamma = 0$	telescoping

Crouzeix–Palencia use $\|\alpha(f)\| \leq \|f\|$; that alone cannot give better than $1 + \sqrt{2}$. Both new proofs use instead that $\alpha(f^n)(T)$ commutes with $f(T)$.

Completely bounded

Conjecture

For every $m \geq 1$ and $F = [f_{ij}] \in M_m(A(\Omega))$, $\|F(T)\| \leq 2 \sup_{z \in \Omega} \|F(z)\|$.

- cb-norm ≤ 2; Paulsen: T similar to an operator with $\overline{W(T)}$ a complete spectral set.
- Known: $1 + \sqrt{2}$ (Crouzeix–Palencia); 2 for discs, weighted shifts (Crouzeix–Greenbaum 2025).
- $M_m(A(\Omega))$ not commutative: both new proofs need the companion to commute with $f(T)$. Crouzeix–Palencia need only $\|\alpha\| \leq 1$.

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Appendix: Γ

$K_W := E_W^*$, $(K_W x)(z) = (I - zW^*)^{-1}x$, $K_0 x \equiv x$. $W = \frac{1}{2}T^*$,
$P = K_W^* K_W \geq I$, $K_W P^{-1} K_W^* =$ projection onto $\operatorname{ran} K_W = N_W^\perp$.

$$\Gamma = (I - K_W P^{-1} K_W^*) K_0 = K_0 - K_W P^{-1} \qquad (K_W^* K_0 = I).$$

$(I - zT)^{-1}(I - zT/2)^{-1} = 2(I - zT)^{-1} - (I - zT/2)^{-1}$:

$$M_R K_W = 2K_{T^*} - K_W, \qquad M_R K_0 = K_{T^*}, \qquad M_R^* K_0 = K_0.$$

$Q = K_W^* K_{T^*}$:

$$\begin{pmatrix} K_W^* \\ K_0^* \end{pmatrix} (M_R + M_R^*) \begin{pmatrix} K_W & K_0 \end{pmatrix} = \begin{pmatrix} 4Q - 2P & Q + I \\ Q + I & 2I \end{pmatrix},$$

$$\Gamma^*(M_R + M_R^*)\Gamma = P^{-1}(4Y - YP - PY)P^{-1}.$$

Appendix: $4Y - YP - PY \geq 0 \Rightarrow \|T\| \leq 2$

$P = \sum_{n \geq 0} 4^{-n} T^{*n} T^n \geq I$, $Y \geq \frac{1}{4} T^* T \geq 0$.

$Px = \alpha x$, $\|x\| = 1$, $\alpha > 2$:

$$0 \leq \langle (4Y - YP - PY)x, x \rangle = (4 - 2\alpha)\langle Yx, x \rangle,$$

so $Yx = 0$, so $Tx = 0$, so $Px = x$. (Approximate eigenvectors, or ultrapower.) So $I \leq P \leq 2I$.

$P = I + \frac{1}{4} T^* P T$:

$$T^* T \leq T^* P T = 4(P - I) \leq 4I.$$

Appendix: the two estimates

$E_nT = TE_n$, $T^*Tx = \kappa^2 x$:

$$\begin{aligned}
\kappa m_n - m_{n+1} &= \operatorname{Re}\langle T^n(\kappa E_n - E_{n+1}T)x, x\rangle \\
&= \kappa(\kappa-1)\|T^{*n}x\|^2 - 2\operatorname{Re}\langle V^*Q^{*n}u, T^{*n}x\rangle \\
&= \kappa(\kappa-1)\left\|T^{*n}x - \frac{V^*Q^{*n}u}{\kappa(\kappa-1)}\right\|^2 - \frac{\|V^*Q^{*n}u\|^2}{\kappa(\kappa-1)} \\
&\geq -\frac{\|u\|^2}{\kappa(\kappa-1)}.
\end{aligned}$$

Multiply by κ^{-n+1}, sum over $n \geq 1$; (m_n) bounded, so the sum telescopes to κm_1.

$\|u\|^2 = \|Q^*VTx\|^2 + \kappa^2 - 2\kappa\operatorname{Re}\langle V^*Q^*VTx, x\rangle$, $\|Q^*VTx\| \leq \kappa$, $2V^*Q^*V = T^* + E_1$:

$$\|u\|^2 \leq 2\kappa^2 - \kappa\operatorname{Re}\langle T^*Tx, x\rangle - \kappa m_1 = 2\kappa^2 - \kappa^3 - \kappa m_1.$$